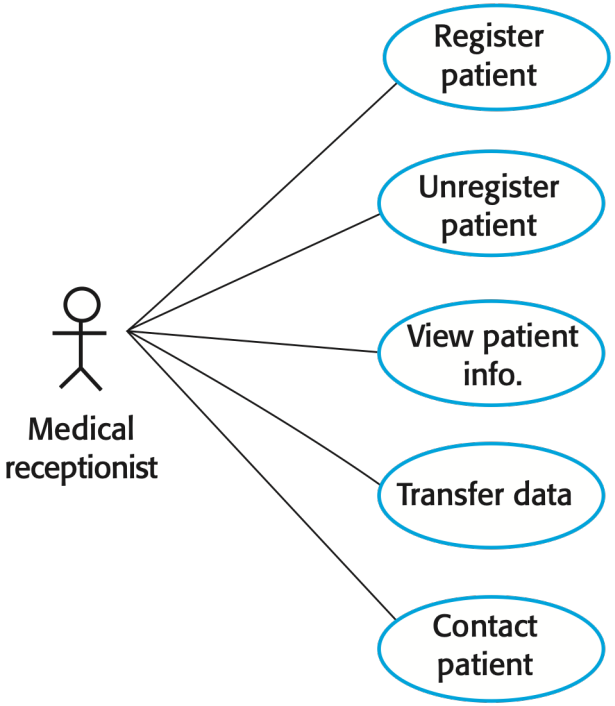
Common question 1: Is it possible to use an arrow instead of a line? (Answers)



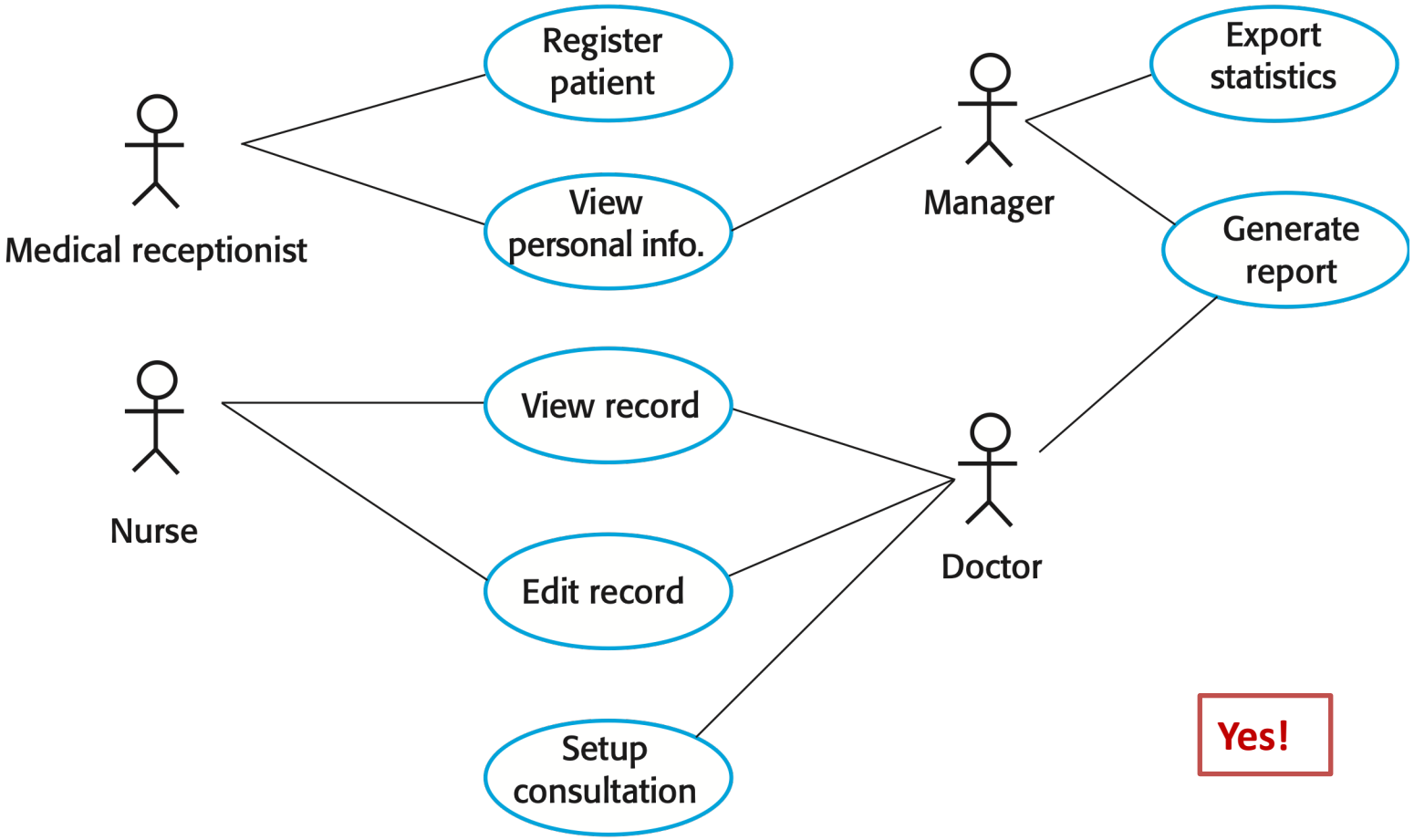
Formally, use case diagrams should use lines without arrows because arrows in the UML indicate the direction of flow of messages.

Common question 2: Is it possible to have multiple use cases within a system? (Answers)

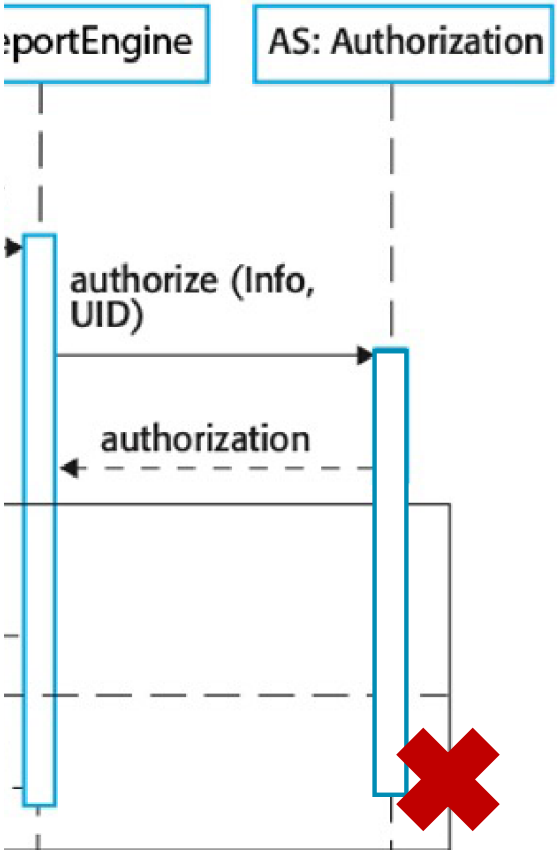


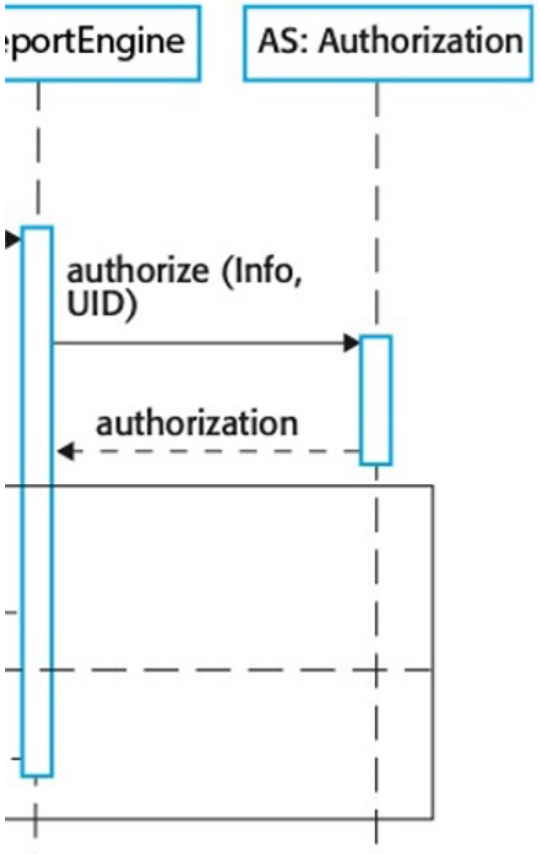
Yes, it is possible to have multiple use cases within a system. Each use case represents a specific interaction or function that the system performs.

Common question 3: Can a system have multiple use cases and multiple actors? (Answers)

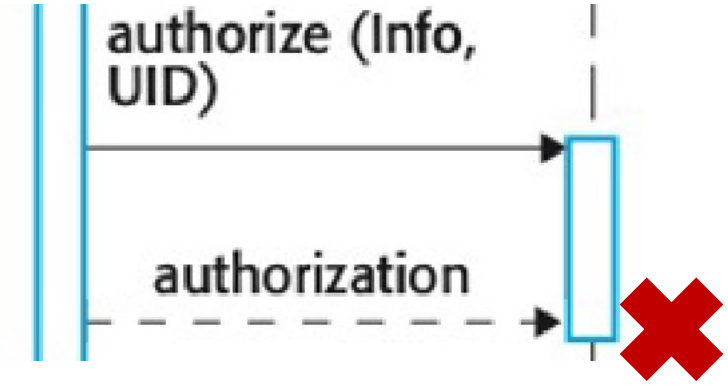


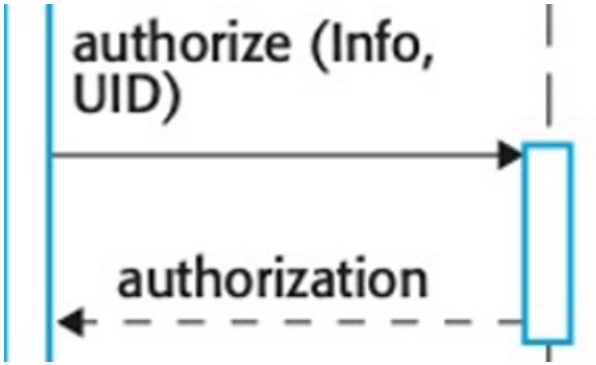
Common Mistake 1: Incorrect Execution occurrence (Answer)



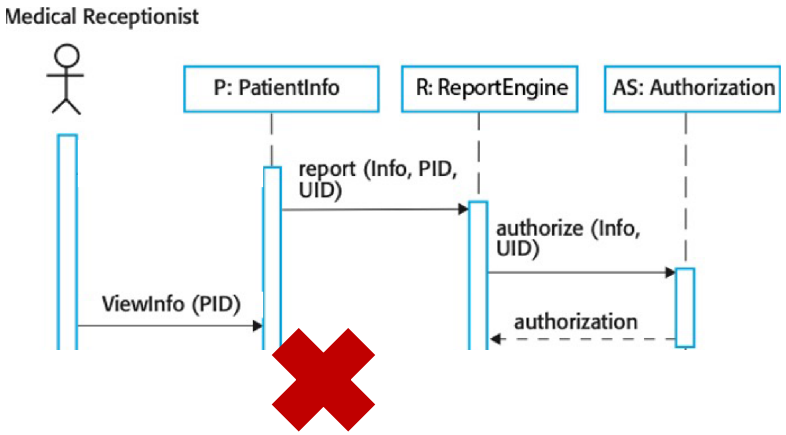


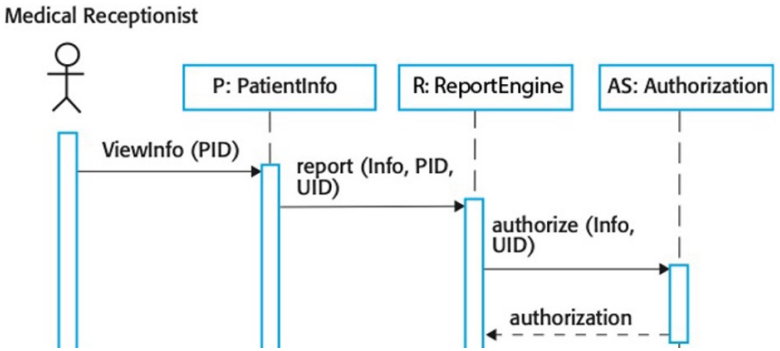
Common Mistake 2: Misplaced Arrows (Answer)



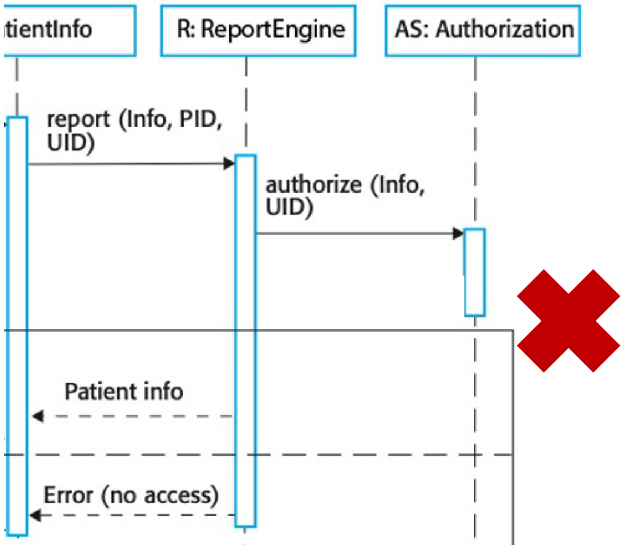


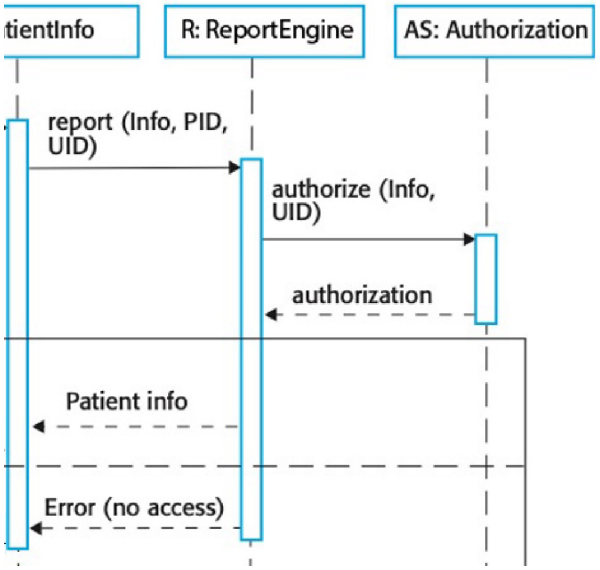
Common Mistake 3: Incorrect Message Order (Answer)



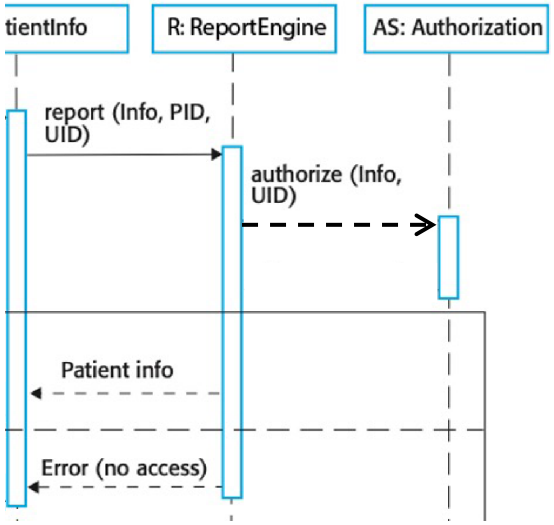


Common Mistake 4: Missing Reply Messages (Answer)

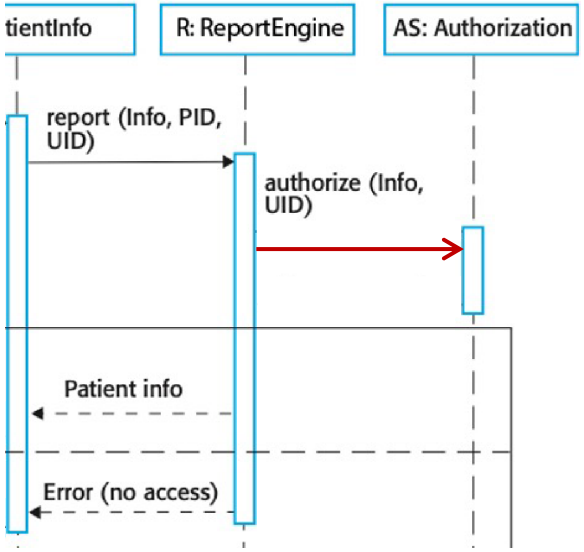




The appropriate use of graphical notation, e.g. asynchronous call, can vary depending on the specific use case (Answer)

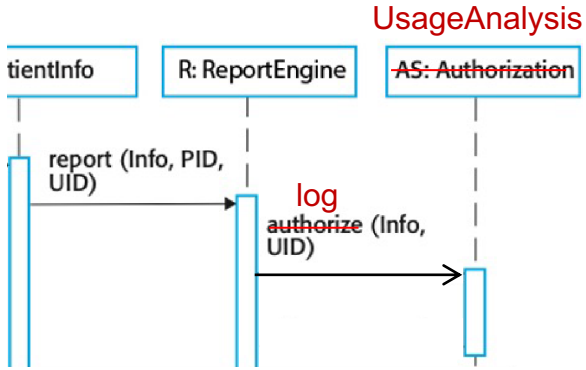


Q1: **Improper use of graphical notation?**



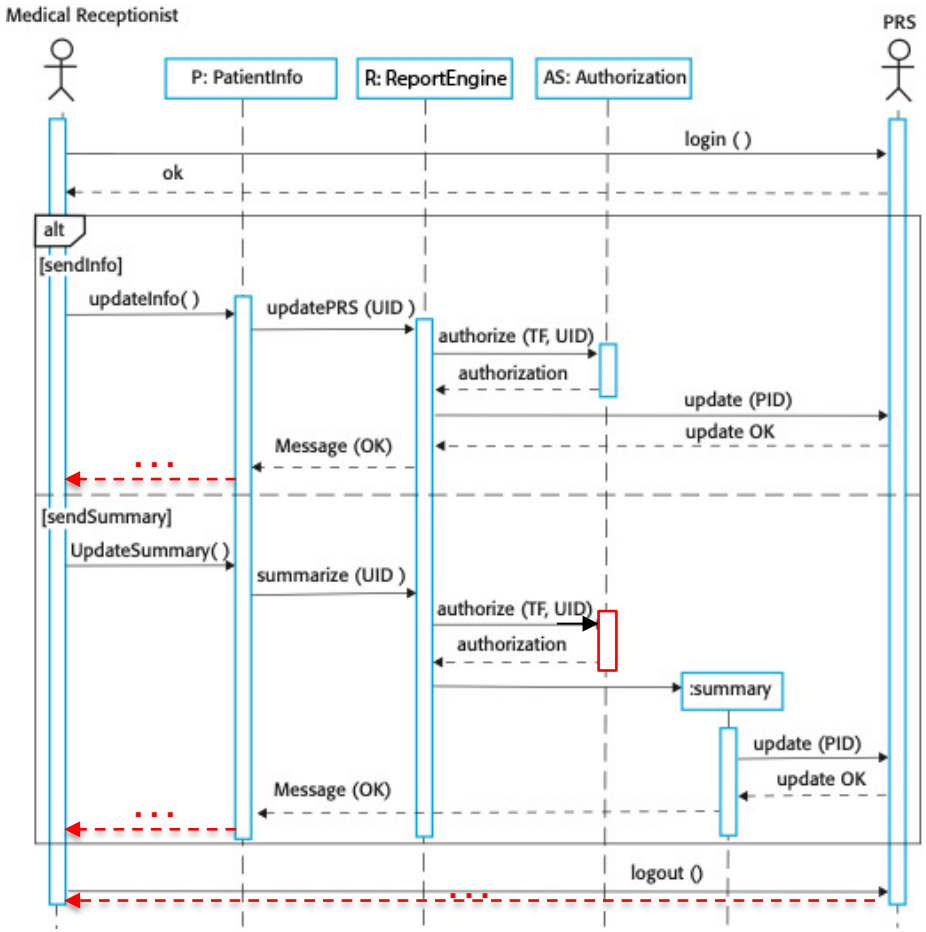
Q2: **Ambiguous message flow?**
In this use case,

- is it appropriate to use an asynchronous message here?
- is it necessary to wait for a return message?

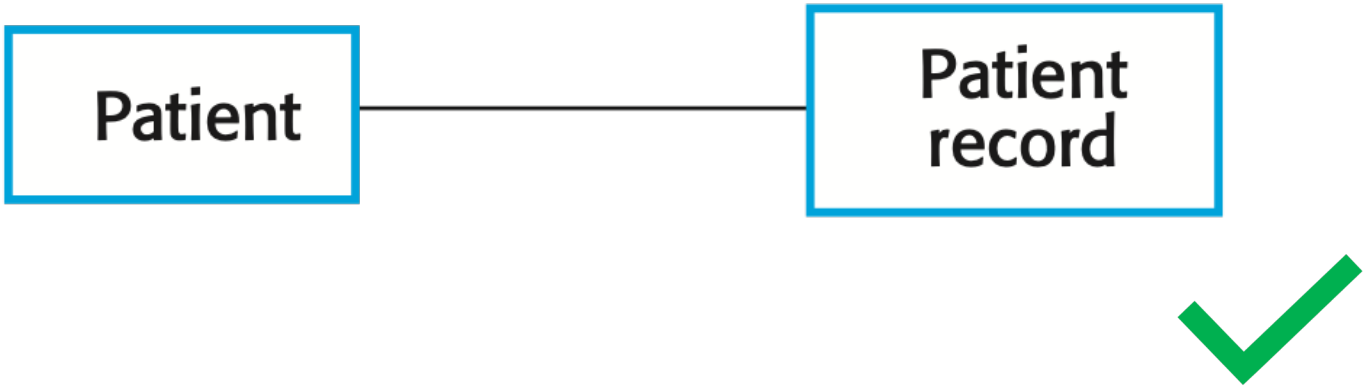


Q3: **Overcomplicated diagram?**

Any UML notation-related improvement(s) from the Sequence Diagram in the previous slide? (Answers)

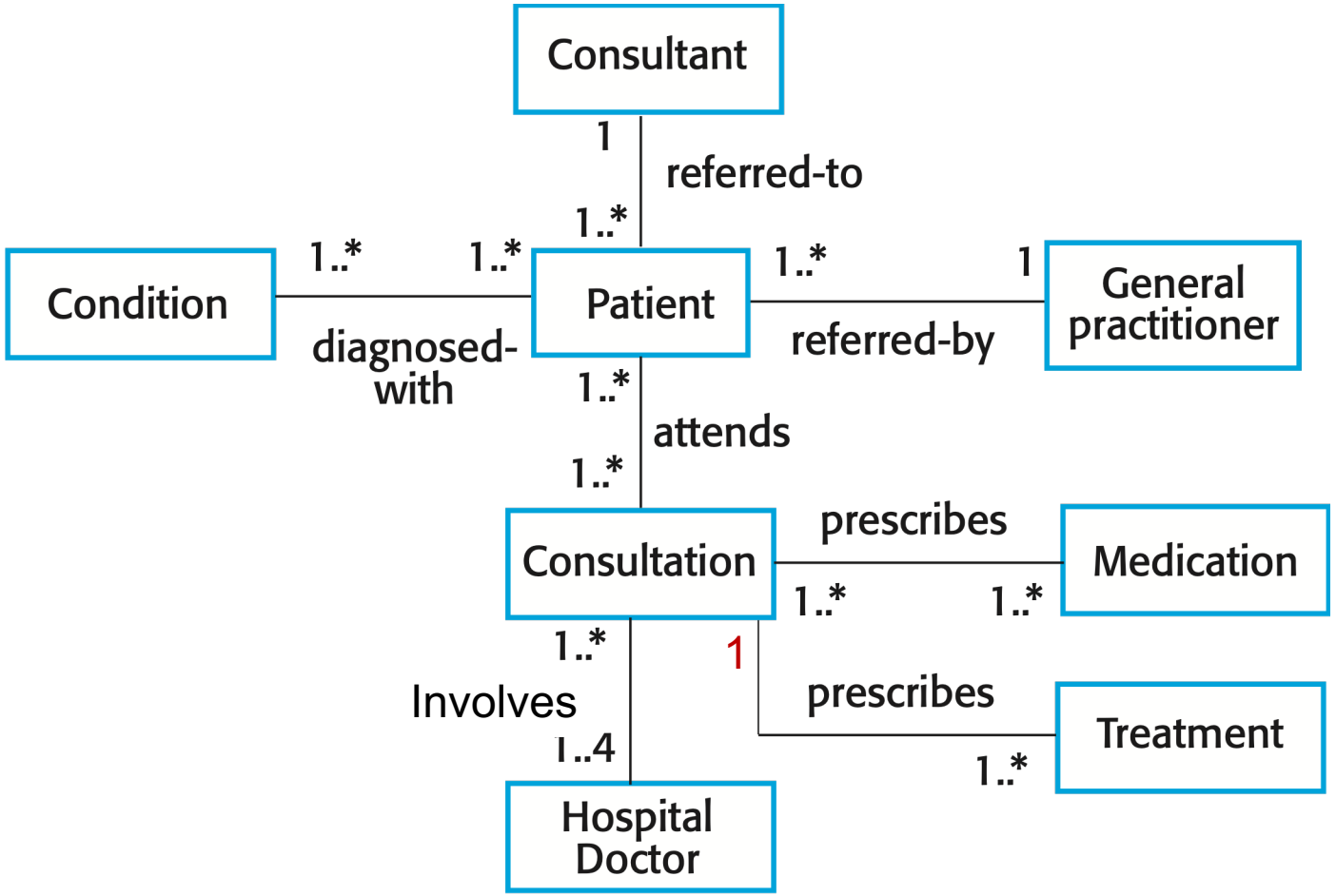


The default multiplicity is “1” (Answer)

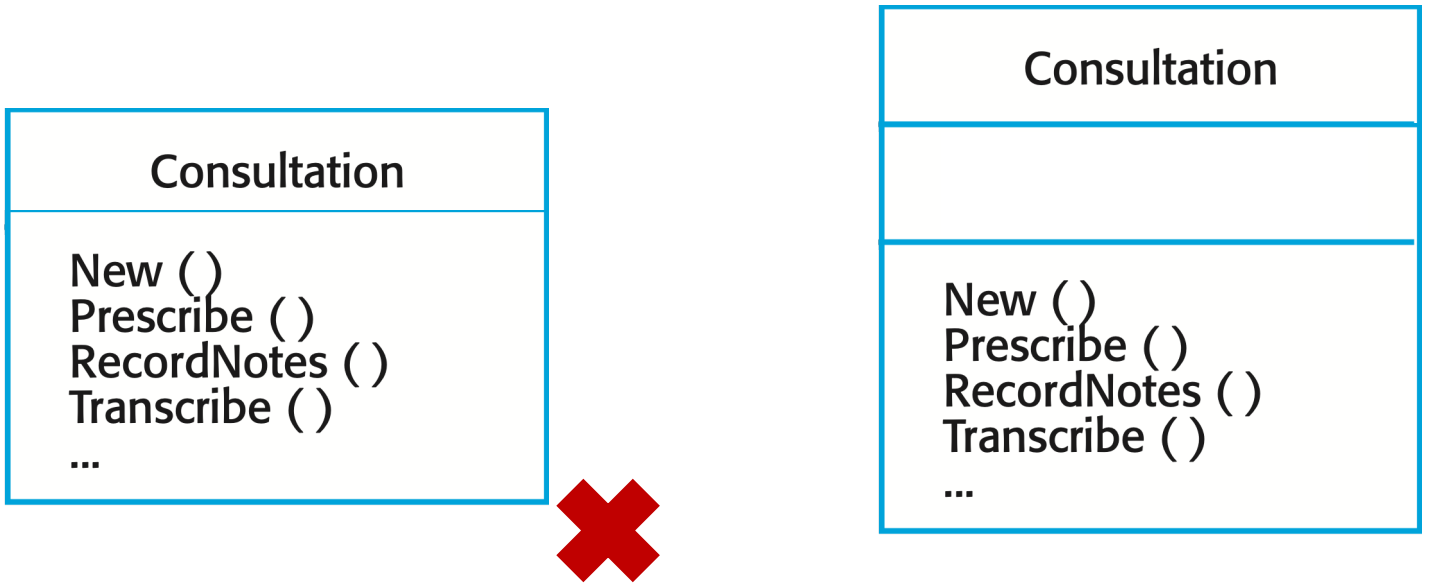


No issue, same meaning as annotated with “1”s in both sides

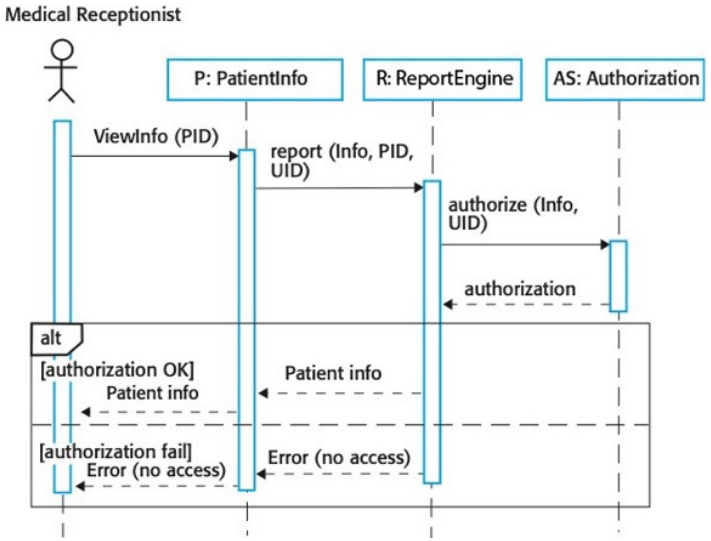
Common Issue 1: Inconsistent (Answers)



Common mistake 1: Improper use of partitions (Answers)



Develop the class diagram from the sequence diagram, step by step. (1) Identify Class

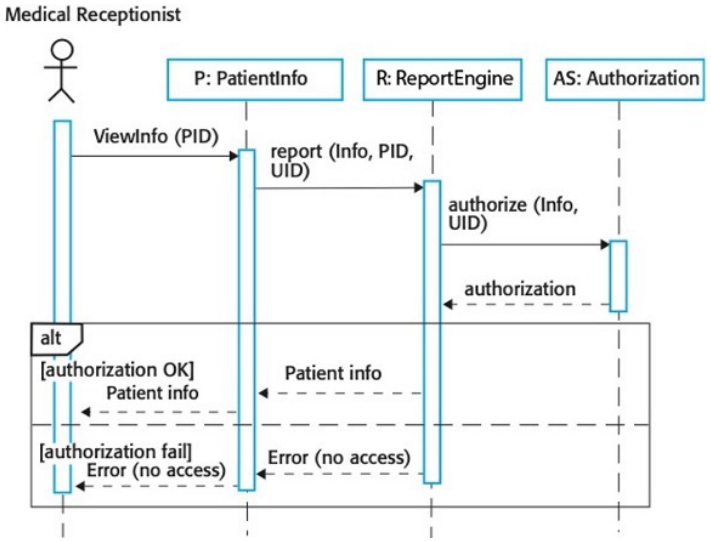


PatientInfo

ReportEngine

Authorization

Develop the class diagram from the sequence diagram, step by step. (2) Identify Method (and types)

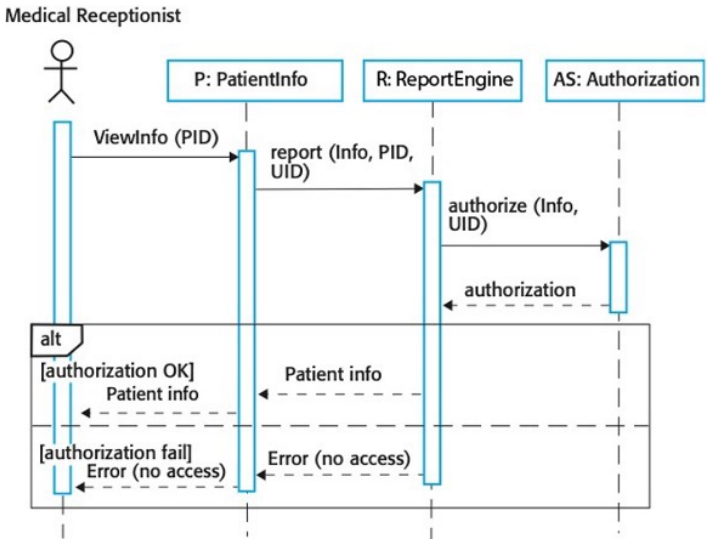


PatientInfo
-viewInfo(PID:int)

ReportEngine
-report(Info:String,PID:int,UID:int)

Authorization
-authorize(Info:String, UID:int)

Develop the class diagram from the sequence diagram, step by step. (3) Identify Attributes (and types)

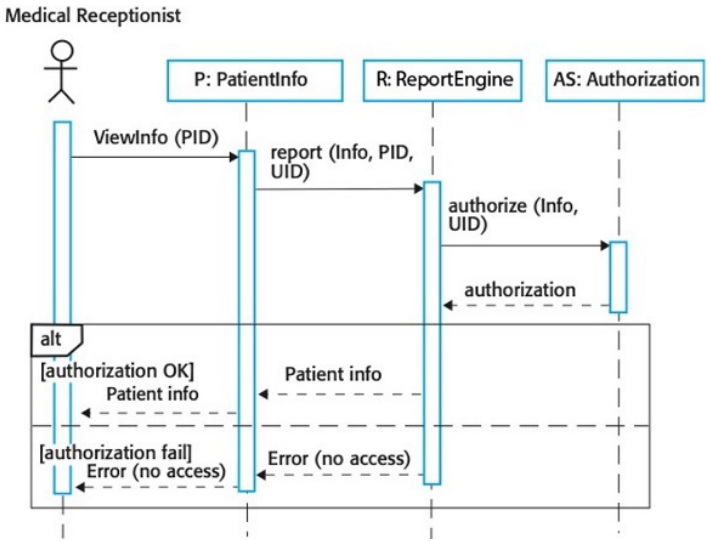


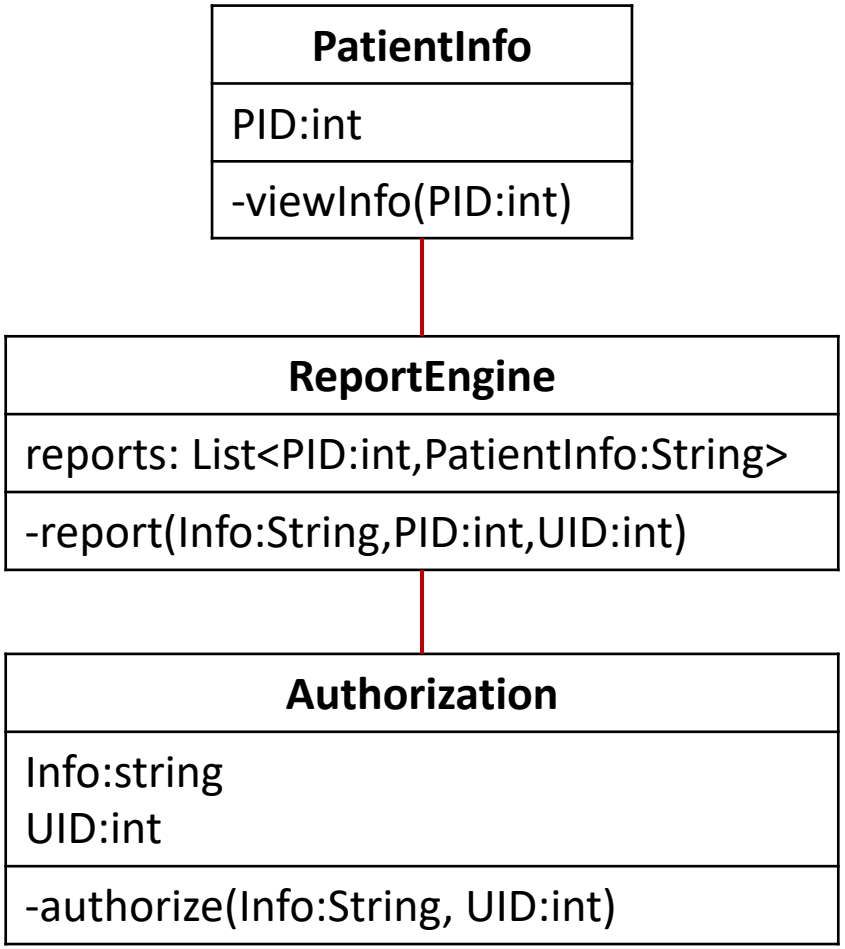
PatientInfo
PID:int
-viewInfo(PID:int)

ReportEngine
reports: List<PID:int,PatientInfo:String>
-report(Info:String,PID:int,UID:int)

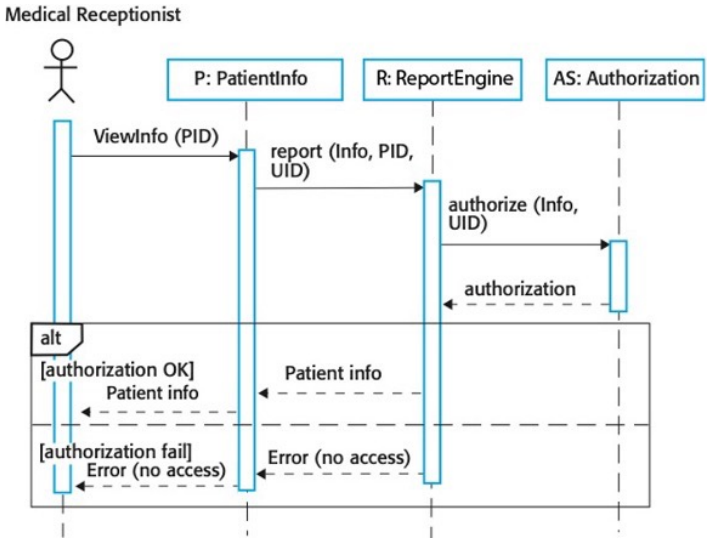
Authorization
Info:string UID:int
-authorize(Info:String, UID:int)

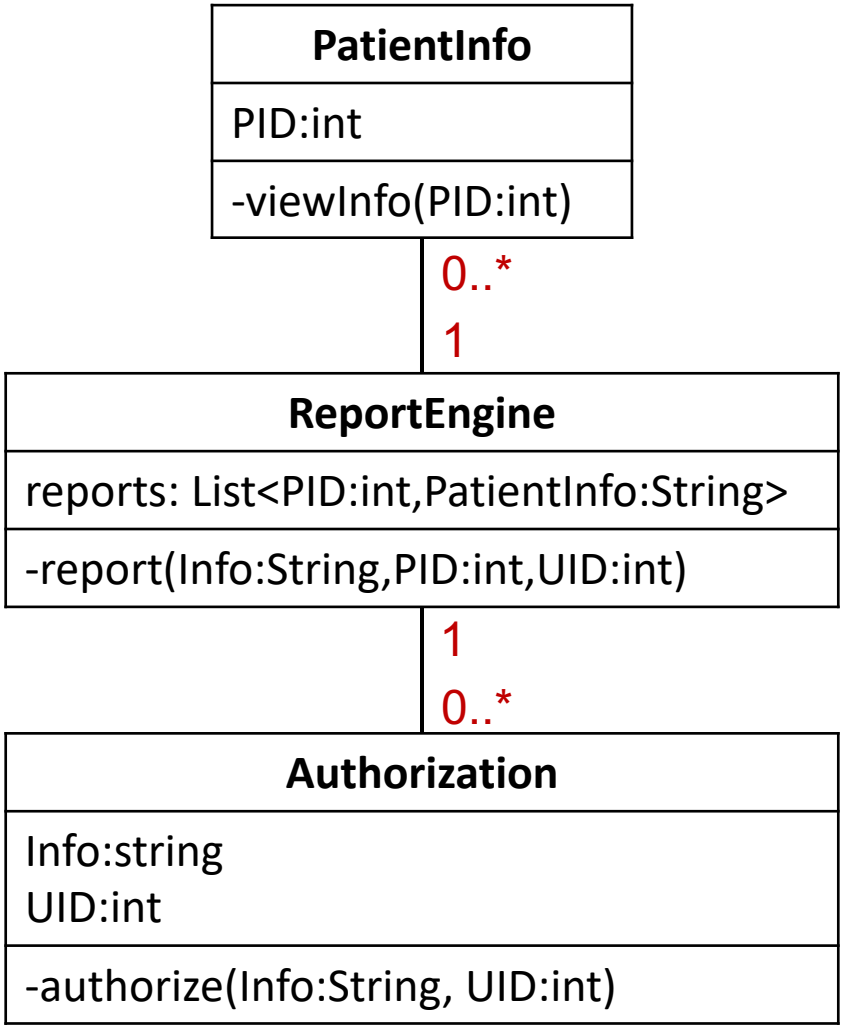
Develop the class diagram from the sequence diagram, step by step. (4) Define Relationship



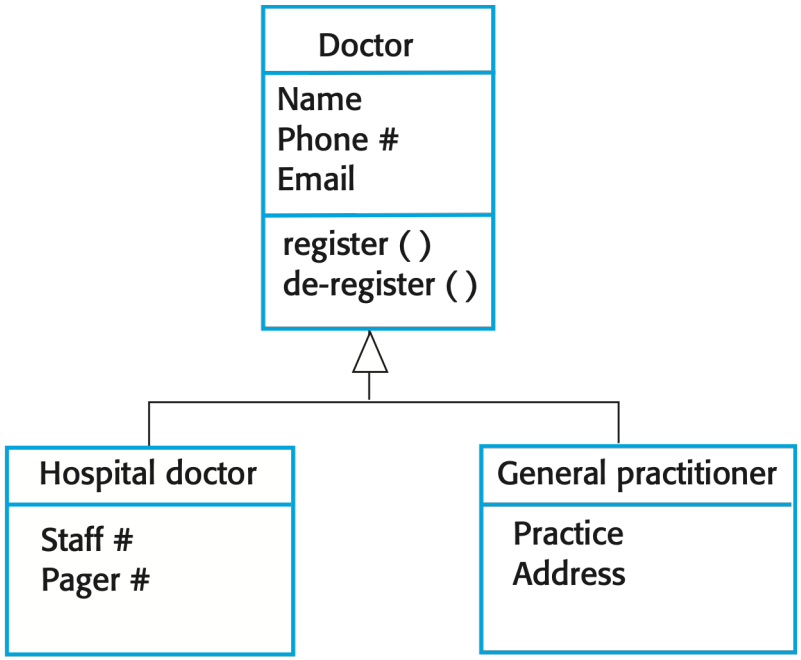


Develop the class diagram from the sequence diagram, step by step. (5) Draw Associations and Multiplicity





How can we maintain the same usage in the UML diagram **without using generalization**? **(Answers)**



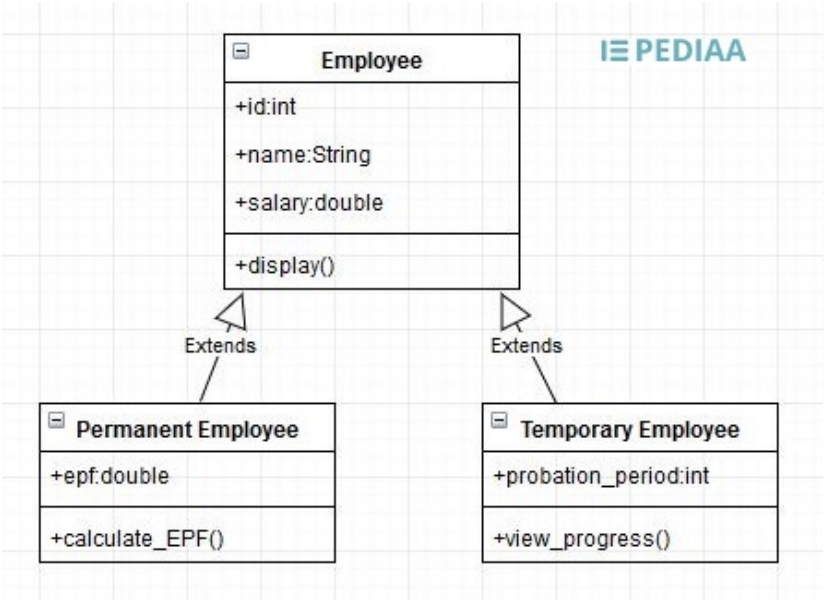
Doctor
Name Phone # Email
register() de-register()

Hospital Doctor
Name Phone # Email Staff # Pager #
register() de-register()

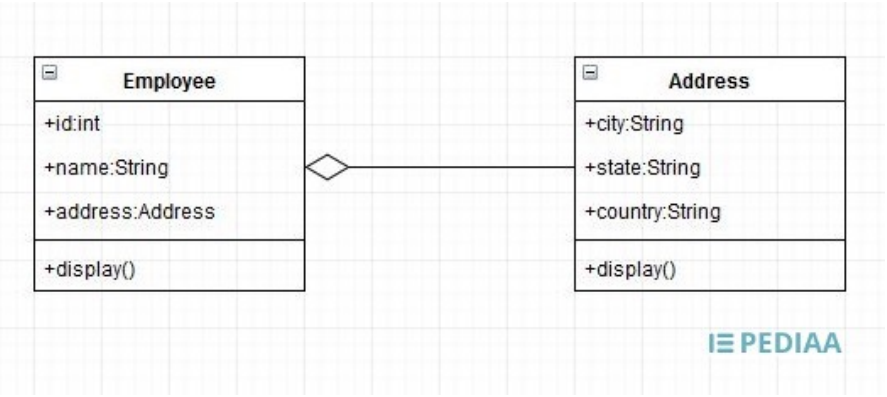
General practitioner
Name Phone # Email Practice Address
register() de-register()

Generalization vs. Aggregation: Understanding the Differences (Answers)

Generalization
"is a" relationship



Aggregation
"has a" relationship



Question

What is the difference considering **implementation**?

(Ref: Pediaa)

Any problem(s)? (Answers)

